

Total No. of Questions :6]

SEAT No. :

P38

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Oct./ TE/ Insem. - 152

T.E. (Civil)

**INFRASTRUCTURE ENGINEERING &
CONSTRUCTION TECHNIQUES
(2015 Course) (Semester - I) (301002)**

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Use of electronic pocket calculator is allowed in the examination.
- 5) Use of cell phone is prohibited in the examination hall.

- Q1)** a) Explain in brief advantages and disadvantages of concrete sleepers. [5]
b) Explain salient features of smart city. [5]

OR

- Q2)** a) Explain coning of wheels and its importance with the help of a neat sketch. [5]
b) Write a short note on metro rails. [5]

- Q3)** a) Discuss in brief different methods to reduce the wearing of rails. [6]
b) Define the following [4]
Equilibrium speed;
Equilibrium cant;
Cant excess;
Cant deficiency.

OR

P.T.O.

Q4) a) Enlist with a neat sketch all elements of a crossing. [6]

b) Explain any 4 types of curves. [4]

Q5) a) List out various dredging systems. Explain any 1 with a neat sketch. [5]

b) Enlist types of barges, explain any two. [5]

OR

Q6) a) What are the various methods of dewatering systems? Explain any one method with a neat sketch. [5]

b) What is a diaphragm wall? State its purposes. [5]

